

Quantitative NEPSE — Complete Site Blueprint

A step-by-step reverse-engineered architecture, feature and data-model document for `quantitativeneipse.com`, written as a build guide.

Compiled 5 August 2026 · Sources: public HTML shell, `/assets/index-B3H9p7K3.js` (583 KB production bundle), `site.webmanifest`, `robots.txt`, and two open endpoints (`/api/health`, `/api/market/health`). No private or authenticated data was accessed.

Scope & ethics note. The site's `robots.txt` carries Cloudflare content signals `search=yes`, `ai-train=no`, `use=reference` and disallows AI crawlers. Everything below is reference-level observation of publicly served files for competitive understanding. It is *not* a licence to copy their code, scrape their API, or reuse their copy text. Rebuild the *ideas*, write your own implementation.

Step 0 — How this blueprint was derived (method you can repeat)

0.1 Fetch the shell. The site is a single-page app; `curl` with a browser User-Agent returns a 4 KB HTML shell (a plain fetch gets HTTP 403 from Cloudflare). The shell exposes: title, meta description, theme bootstrapping, analytics ID, and the asset manifest.

0.2 Read the module preloads. `<link rel="modulepreload">` tags name the vendor chunks — `echarts`, `recharts`, `lightweight-charts` — which reveals the charting stack before a single page renders.

0.3 Download the main bundle and grep it. A Vite production build keeps route definitions, endpoint literals and all UI copy as plain strings. Three greps give you the whole product:

```
grep -oE 'path:"[^"]{1,60}"' index.js | sort -u # routes
grep -oE '"/api/[A-Za-z0-9_/.:}{?=&-]+' index.js | sort -u # endpoints
grep -oE '"[A-Z][a-z][^"]{15,90}"' index.js | sort -u # UI copy / tooltips
```

0.4 Probe the open endpoints. `/api/health` and `/api/market/health` answer without auth and leak the data source, the freshness model and the entire market-regime scoring schema.

0.5 Read the manifest and robots. Confirms PWA install, brand colours, and the crawler policy.

Step 1 — Technology stack

Layer	Observed implementation	Evidence
Frontend	React SPA built with Vite (hashed ESM chunks, <code>crossorigin</code> module entry). React Router v6+ for routing.	<code>/assets/index-B3H9p7K3.js</code> ; router warning strings ("Index routes must not have child routes")
Styling	Tailwind CSS. Dark theme is the default; light only if explicitly chosen. Theme persisted in <code>localStorage["qnepse-theme-v1"]</code> with a legacy fallback key, applied by an inline pre-paint script to avoid a flash.	Inline head script; class strings like <code>flex min-h-screen flex-col bg-background</code>
Typography	Inter (variable, 400–700) for UI, JetBrains Mono for numerics — served from Google Fonts with preconnect.	Font <code><link></code>
Charts	Four libraries, each for a different job: • ECharts — Sankey, rotation maps, heat/bubble • Recharts — small statistical panels, trends • Lightweight Charts — fast embedded price series • TradingView Advanced Charts — the full <code>/technical-chart</code> terminal, fed by a UDF datafeed	<code>modulepreloads;</code> <code>/charting_library/charting_library.standalone.js</code> , <code>/datafeeds/udf/dist/bundle.js</code> , endpoint <code>/api/udf</code>
Auth	Clerk , full-featured: sign-in/up, session tasks, MFA setup, password reset, organizations + org switcher, user profile, redirect callbacks. Even the Metamask/Web3 sign-in path is bundled (likely unused default).	Strings <code>TaskSetupMFA</code> , <code>OrganizationSwitcher</code> , <code>AuthenticateWithRedirectCallback</code> , <code>SignInWithMetamask</code>
Billing	Clerk Billing with Stripe — a <code>PricingTable</code> component, Stripe payment element contexts, and plan/plan-period/immediate-plan-change logic are all compiled in. A paid tier exists or is imminent.	<code>"PricingTable"</code> , <code>StripeUtilsContext</code> , <code>PaymentElementContext</code> , <code>isImmediatePlanChange</code> , <code>planPeriodStart</code>
Backend	A JSON API under <code>/api/*</code> , same origin. Analytical store is DuckDB , a single file on a Linux host, populated by a batch	<code>/api/health</code> → <code>"dataSource":"duckdb"</code> , path <code>/root/projects/quantitative_nepse/daily-market-analysis/data/duckdb/nepse.duckdb</code> , <code>"cacheTtlSeconds":120</code>

	pipeline. Responses carry a 120-second cache TTL.	
Discussion store	A separate database, optional — the app degrades gracefully when it is absent ("Community pulse needs the discussion database configured on the API").	Error copy
Infrastructure	Cloudflare in front (bot management, managed robots block, <code>cf-cache-status: DYNAMIC</code> on API). PWA-installable (standalone display, brand colour <code>#0c1222</code>).	Response headers; <code>site.webmanifest</code>
Analytics	Google Analytics 4, property <code>G-PDD96PX18W</code> , loaded async in head.	HTML shell
SEO / social	Canonical URL, full Open Graph set with 1200×630 image and alt text, Twitter <code>summary_large_image</code> , SVG favicon, Apple touch icon.	HTML shell

Architectural summary in one line: a static React SPA on a CDN, talking to a thin read-only JSON API over a nightly-batched DuckDB analytical file, with authentication and billing fully delegated to Clerk+Stripe.

Step 2 — Route map (13 routes)

Route	Page name in UI	Purpose
/	Dashboard	Market Insights home — pulse, health score, breadth, whale summary, sector rotation, macro backdrop
/live	Live market	Live quote board during TMS session hours; falls back to last stored bars off-session
/floorsheet	Order flow	Broker-level flow analytics — the flagship module
/stock	Stock Profile	Symbol picker + personal profile watchlist
/stock/:ticker	Stock Profile	"Price, fundamentals, technicals & order flow — one screen"
/fundamentals	Fundamental Insights	Eight-panel sector-aware fundamental model
/screens	Screens	Technical & relative-strength screens, style tabs, filters
/technical-chart	Technical chart	TradingView Advanced Charts terminal
/macro	Macroeconomics	NRB macroeconomic indicator dashboard
/watchlist	Watchlist	Stock, sector and broker watchlists in one place
/discussion	Guf Gaaf	Social feed with cashtags and bull/bear voting
/discussion/user/:userId	User profile	Public posts, followers/following, optional public watchlist
/charting_library/	—	Static TradingView library assets (licensed, self-hosted)

Step 3 — API surface (34 endpoints)

3.1 Market state

Endpoint	Returns
/api/health	Data-source name and hint, <code>asOf</code> , <code>pricesAsOf</code> , <code>indicesAsOf</code> , <code>cacheTtlSeconds</code> , and a <code>dataUpdateState</code> / <code>Label</code> / <code>Hint</code> triad driving the freshness banner
/api/market/health	Composite score 0–100, <code>regime</code> + <code>regimeBadge</code> , and a <code>regimeDetail</code> object: alignment, <code>macroTone</code> , <code>marketTone</code> , <code>fundamentalsTone</code> , one theme string per pillar, plus scored <code>bullFactors</code> / <code>bearFactors</code> arrays
/api/market/live	Full live quote board
/api/market/live?fields=pulse	Slim pulse payload for the dashboard header (same route, projection param — a good pattern)
/api/market/whale-summary	Large-participant flow digest
/api/market/top-brokers-turnover	Broker turnover leaderboard

3.2 Price & reference data

Endpoint	Returns
/api/data/stock-ohlc	Per-symbol OHLC series
/api/data/indices-ohlc	Index OHLC + monetary turnover volume bars
/api/data/turnover	Turnover series / concentration inputs
/api/data/stock-flags	Style and status tags per symbol (Growth, Pre-Revenue, etc.)
/api/data/stock-profile-page	Composite payload for the whole Stock Profile page
/api/data/stock-profile-core	Core price/technical block
/api/data/stock-profile-fundamentals	Fundamentals block
/api/tickers	Master symbol list for search (Ctrl+K)
/api/fundamentals/tickers	Symbols that actually have published statements
/api/udf	TradingView UDF datafeed (config/symbols/history/marks)

3.3 Floorsheet insights

Endpoint	Returns
/api/floorsheet-insights/brokers	Broker directory
/api/floorsheet-insights/top-brokers	Ranked by 30-day average gross turnover (buy + sell)
/api/floorsheet-insights/broker-stocks	What one broker traded, by symbol
/api/floorsheet-insights/ticker-brokers	Who traded one symbol
/api/floorsheet-insights/ticker-broker-sankey	Broker→broker / broker→ticker net-buy flow graph
/api/floorsheet-insights/market-breadth	Breadth % (net buyers) across the board
/api/floorsheet-insights/trading-dates	Valid session dates for the slider/date picker

3.4 Analytics & macro

Endpoint	Returns
/api/insights-dashboard	Dashboard aggregate
/api/insights-dashboard/signals	Screen/signal results
/api/macro/dashboard	NRB macro indicator set

3.5 Discussion (Guf Gaaf)

Endpoint	Returns
/api/discussion/feed	Main timeline
/api/discussion/posts	Create / read posts
/api/discussion/insights	Community pulse — bullish/bearish tallies, most-mentioned symbols
/api/discussion/media?kind=post	Image upload (sign-in required)
/api/discussion/notifications	Notification list
/api/discussion/notifications/read	Mark read
/api/discussion/notifications/unread-count	Badge counter (polled)
/api/discussion/users/me/watchlist	User's public/private watchlist

Step 4 — Module blueprints (feature by feature)

4.1 Dashboard — Market Insights

The landing page composes six blocks, each backed by its own aggregate so a slow one never blocks the page:

1. **Market pulse** — NEPSE session state, live index, turnover, up/down counts. Explicit indicator that the market is not in an active TMS session.
2. **Market health score** — 0–100 composite mapped to a regime badge (observed: score 36 → "MODERATELY BEARISH"). Decomposed into three pillars with tone and theme:
 - **macro** constructive — "easing financial conditions"
 - **market** headwind — "narrow breadth"
 - **fundamentals** headwind — "weak earnings quality"
 - plus an **alignment** field (mixed/aligned) and per-factor scored bull/bear arrays, e.g. `{key:"lending_rate", label:"Lending rate", score:...}`
3. **Market breadth** — advance/decline, breadth vs NEPSE, stocks above MA by sector.
4. **Market cycle** — a model-assigned phase label plus a 0–100 model confidence for that phase.
5. **Sector rotation map** and **relative strength map** — ECharts quadrant/bubble views; weighted mode scales each index by turnover share, unweighted shows plain price return.
6. **Macro backdrop** — collapsible summary that links to [/macro](#).

Design lesson: the health score is never shown alone. Every score ships with its drivers, tones and themes, so the number is auditable by the reader.

4.2 Live market

- Live quote board with 20-session % change and previous-close % change computed from live candles.
- Session awareness is first-class: green = market hours, live quotes active; the EOD database updates only after close.
- Off-session it explicitly says it is showing the last stored bars for today rather than silently showing stale data.

4.3 Floorsheet / Order flow & smart money **flagship**

This is the module the whole product is marketed on ("whale broker flow"). Components:

Component	Detail
Top brokers	Ranked by 30-day average gross turnover (buy + sell) — deliberately not a single session, with the copy "Top brokers aren't crowned on one lucky session."
Top Brokers: Buy vs Sell	Paired bar view of each broker's buy and sell side
Largest net buyers / net sellers	Ranked lists with a configurable "number of ranked lines shown"
Cumulative net buy	Running net-buy curve per broker and per ticker over the selected window
Net buy flow (Sankey)	ECharts Sankey of broker→ticker net flow for a chosen symbol and window
Concentration metrics	Top-5 buyer concentration, Top-5 seller concentration, turnover concentration, net-flow balance — the accumulation-vs-distribution read
Market breadth (flow)	Breadth % measured as net buyers, not price advances
Conviction leaders · 5D	Five-day persistent-accumulation leaderboard
Time control	A lookback slider across trading history ("Older ← drag → Latest") plus "Play animation from first date" — animated bubbles where <i>size is habit and height is today</i>
Watchlist integration	Brokers themselves can be watchlisted

4.4 Stock Profile — `/stock/:ticker`

Positioned as "Price, fundamentals, technicals & order flow — one screen." Sections observed:

- **Price** — OHLC candlesticks, latest session close from the live series, % change vs previous close, % change over the last 20 *trading sessions* (explicitly not calendar days).
- **Technical** — RSI, MA10/MA20/MA50/MA200, Beta 6M, trend/price-position labels, "indicator batch date can lag the live chart by a session" disclosure.
- **Floorsheet Signal** — the order-flow verdict for this symbol, with conviction drivers.
- **Valuation Signal** — Undervalued / Fairly Valued / Overvalued.
- **Peer rank** — percentile within sector, "percentile shows how many peers you beat on this metric", ranked against each peer's *latest published statement period*.
- **Growth & dividends** — dividend history by year, latest cash/bonus/total, payout as % of face value (Rs 100), 5-year average, payout consistency.
- **Financial statements** — published statements viewer.
- **Order flow** — ticker-broker table and Sankey embedded.
- **Discussion** — recent Guf Gaaf posts mentioning the symbol.
- **Advanced chart** ↗ — hand-off to the TradingView terminal.

4.5 Fundamental Insights — the eight-panel model

The most elaborate module. It is **sector-aware**: banks, insurers, hydropower and general corporates each get different metrics and different pass/fail floors.

Panel 1 — Two-layer quality gate

- **Layer 1 — Regulatory Gate Checks.** Sector regulatory floors: capital, NPL, solvency, profitability. "Equity returns must stay positive to clear this gate" — a loss-making company fails outright. Passing all gates earns the badge *Sound*; failing steps the grade down ("Gate penalty: grade stepped down ...").
- **Layer 2 — Absolute Quality Anchors.** Scored against absolute thresholds, not peers, so a whole weak sector cannot look good by relative comparison.

Panel 2 — Earnings Power & Growth

YoY EPS %, YoY Earnings %, YoY Revenue %, QoQ Earnings, annualised EPS, plus a regression trend line. Growth style tag requires at least 4 years of history — otherwise it says so rather than guessing.

Panel 3 — DuPont ROE decomposition

ROE broken into **Net Profit Margin × Asset Utilization × Equity Multiplier (Leverage)**, charted as factors over time, with a reconstructed-ROE reconciliation check and narrative labels ("Productivity story:", "Balanced drivers:", "Mixed signals: earnings and operating surplus moving in different directions").

Panel 4 — Asset Quality (BFI only)

NPL ratio, provision coverage, impairment as % of loans, with a trend verdict — "Asset quality improving: NPL trending down and coverage holding. No deterioration flag." Each metric carries a plain-English tooltip.

Panel 5 — Capital & Funding (BFI only)

CAR (Capital Fund / RWA, NRB min 11%), CET1, CDR, cost of funds, NI spread, base lending rate, deposit growth %, loan growth %, reserve & surplus / paid-up capital. NRB regulatory thresholds are drawn as reference lines on the charts.

Panel 6 — Relative Valuation

P/E and P/B versus sector median; peer quartile bands labelled *Top 25% — Outstanding, Upper Half — Above Median, Lower Half — Below Median, Bottom 25% — Lagging Peers*. The docs are careful to note "peer quartile (not fiscal quarter)" and that the quartile is the average of per-metric percentiles within the sector.

Panel 7 — Absolute / Intrinsic Valuation

Justified P/B = $(\text{ROE} - g) / (r - g)$ — Gordon Growth, cross-checked against an earnings-yield view.

User-editable inputs: Cost of Equity (r), Risk-Free Rate (for earnings yield), Sustainable Growth Rate (g), Return on Equity (ROE).

Output: a **sensitivity grid** across ROE and g ; green when justified P/B exceeds current market P/B.

Verdict rule: *below intrinsic* → *Undervalued*; *within +10%* → *Fair value*; *above* → *Expensive*.

Explicit degradation: "Insufficient data for justified P/B", "No P/B + ROE pair available for this sector/ticker, so no intrinsic signal", "Estimate withheld".

Panel 8 — Dividend Durability

- Payout ratio trend, EPS vs DPS coverage analysis.
- **Dividend Capacity Index** — binary health checks supporting the *ability* to pay, distinguished sharply from historical habit: "This is balance-sheet/earnings capacity — not historical payout consistency."

- Forward capacity band filter; warning when "estimated capacity is under 5% of face value".
- Dividend reliability: how often the company paid across its history; 5-year average total payout.

Sector-specific metric sets observed

Sector	Metrics
Commercial banks / Development banks / Microfinance	CAR, CET1, NPL, provision coverage, CDR, cost of funds, NI spread, base lending rate, net interest margin, operating expenses / operating income, financial expenses / operating income, deposit & loan growth
Life & Non-Life Insurance	Gross Premium / Total Assets, Net Premium / Gross Premium, Net Profit / Gross Premium, Net Profit Margin (Investment), Agent Commission / Gross Premium, Unpaid Claims / Policies, Total Inforce Policies, solvency cushion vs regulatory floor
Hydropower	Royalty / Revenue from Electricity, Capital Work in Progress (CWIP % of assets), Pre-Revenue/Under-Construction flag, Loans & LT Liabilities / Total Assets, positive-equity-return check
Manufacturing, Hotels & Tourism, general corporates	Gross / Operating / Net Profit Margin, Inventory Turnover, Operating Income / Total Assets (annualised), leverage vs equity and vs assets, Reserve & Surplus / Paid-Up Capital, Net Worth Per Share

Panels 4 and 5 are hidden for non-BFI sectors: "Sector-specific panels (Asset Quality / Capital) are available for BFI sectors only."

4.6 Screens

- Style tabs (e.g. *Trend & momentum*, *Technical & relative strength*, growth/value style tags).
- Cross-filters: reported sector within the active style tab; peer-relative valuation or ROE snapshot tags ("Cheap vs peers"); dividend-capacity band.
- Compound filtering — "narrow the list to names that also carry another style tag in addition to the active tab".
- Result rows carry both fundamental and flow context, so a screen hit is immediately actionable.
- Historical window control for cumulative return lines — "longer windows smooth short-term noise".

4.7 Technical chart

Self-hosted TradingView Advanced Charts (licensed) driven by a UDF datafeed at `/api/udf`. The app fails gracefully with "TradingView charting library not installed" if the licensed folder is missing — meaning the repo ships without it and the deploy step copies it in.

4.8 Macroeconomics

NRB (Nepal Rastra Bank) indicator dashboard: base rate, lending rate, deposit rate, interest spread rate, and related series. Feeds the macro pillar of the market-health score, which is why "easing financial conditions" can appear as a constructive theme on the dashboard.

4.9 Watchlists

Three independent watchlist types — **stock**, **sector** and **broker** — each with add/remove/clear, plus a separate lightweight Stock-Profile watchlist for one-click symbol switching. A user's stock watchlist can be marked *Public* — *visible on your profile* or *Private* — *only you can see this*.

4.10 Guf Gaaf — the social layer

Feature	Detail
Posting	Free text with cashtags : "Type \$ and a ticker (e.g. \$NABIL) to tag a stock." Image upload requires sign-in.
Sentiment voting	Every post can be marked bullish or bearish by other users; voter lists are viewable.
Engagement	Likes, comments, reposts; per-post comment counts.
Social graph	Follow / followers, public user profiles, optional public watchlist on profile. Name and photo come from the Clerk account menu.
Notifications	Feed + unread-count badge.
Community pulse	Aggregations back into market data: <i>Most popular</i> , <i>Most talked about</i> , <i>Most bullish</i> , <i>Most bearish</i> , <i>Stocks mentioned in posts</i> , total bullish/bearish votes on tagged posts — sorted by likes, bullish, bearish, comments and reposts.
Read/write split	Reading is public; posting, commenting and voting require sign-in.

Step 5 — Cross-cutting design patterns worth copying

5.1 Data-freshness state machine. Instead of a spinner, the API returns an explicit state (`updated` / `refreshing` / `stale`) with label and hint, and the UI renders a coloured banner: green = DB up to date, green = live market hours, red = EOD DB has not caught up after close, plus "Post-market update in progress" and "Closing bell update". Users always know exactly how old the number is.

5.2 Every metric has a one-line plain-English tooltip. The bulk of the string table is explanations like "How fully deposits are lent out. Too high stresses liquidity; too low under-uses funding." This is what converts a data dump into a research product — the tagline is literally "From confusion to conviction".

5.3 Scores always expose drivers. "Show conviction drivers" / "Show grade drivers" — each driver adds or subtracts from a running conviction score that sets the bucket. Drivers cover trend structure, momentum, participation, positioning and valuation posture. No black-box number.

5.4 Honest absence of data. Dozens of distinct empty states rather than blank charts: "Insufficient peer data to rank this ticker", "Growth history is too short for a Growth style tag (needs ≥ 4 years)", "Estimate withheld", "No multi-year history for this stock yet". Refusing to compute is treated as a feature.

5.5 Personality in loading copy. "Our quant gnomes are updating today's session", "Loading OHLC — open, high, low, and patience", "Checking if your picks are bullish or just hopeful...", "Almost there — quicker than a T+2 settlement". Cheap to write, disproportionately memorable.

5.6 Clear positioning guardrail. "Research only · Not investment advice" is a permanent chrome element. There is no portfolio, no P&L, no broker order routing — the product is deliberately read-only, which keeps it out of regulatory scope.

5.7 Absolute gates before relative ranking. Layer 1 regulatory floors run before any peer percentile, so the best of a bad sector cannot be dressed up as quality. This is a genuinely strong modelling idea.

5.8 Composite endpoint per page, slim projection for widgets. `/api/data/stock-profile-page` serves the whole page in one round trip, while `/api/market/live?fields=pulse` trims the same route for a header widget. Fewer requests, one cache key per view.

5.9 Keyboard-first search. Ctrl+K symbol search wired globally; separate scoped searches for tickers, brokers and sectors.

5.10 Chart interaction discipline. "Click to interact with chart", "Chart active — click outside to scroll page", "Scroll or pinch to zoom · drag to pan", reset-zoom button. Solves the classic mobile problem of charts hijacking page scroll.

Step 6 — Data pipeline implied by the observations

NEPSE sources	Batch (nightly / post-close)	Serving
floorsheet (per trade)	ETL → DuckDB single file nepse.duckdb + derived tables: • broker daily net flow • rolling 30d broker turnover • technical indicator batch • sector percentile tables • fundamental gate/grade scores	JSON API → React SPA (120 s cache) (Cloudflare CDN)
EOD prices / indices		
published financials		
NRB macro indicators		
live TMS quotes (session hours only)		/api/market/live (overlay on EOD)
discussion posts	separate DB (optional)	/api/discussion/*

- **DuckDB as the analytical store** is the notable choice: a single embedded columnar file, ideal for the wide GROUP-BY scans that broker-flow analytics need, with no database server to run.
- **Indicators are batched, not live** — the UI admits the "indicator batch date can lag the live chart by a session".
- **Live quotes are an overlay** on the EOD store during TMS hours only; outside those hours the last stored bars are shown, labelled as such.
- **120-second cache TTL** on API responses, with Cloudflare marking them `DYNAMIC` (not edge-cached), so the TTL is application-level.

Step 7 — Rebuild checklist, in dependency order

1 Ingest. Daily floorsheet, EOD prices, index OHLC, published financial statements, NRB macro series. Store raw; never mutate raw.

2 Derive. Build per-day broker aggregates first, then merge across ranges — never a single wide GROUP BY over the raw floorsheet. Precompute rolling 30-day broker turnover, per-symbol net flow, technical indicator batches and sector percentile tables.

3 Health contract. Ship a `/health` that reports `asOf`, per-domain freshness and an explicit update state. Build the UI banner against it on day one — every later debugging session depends on it.

4 Reference endpoints. Ticker master, fundamentals-available tickers, broker directory, trading dates.

5 Flow module. Top brokers by 30-day gross turnover → net buyers/sellers → cumulative net-buy → Sankey → concentration metrics → date slider and animation. Concentration (top-5 buyer vs top-5 seller share) is what turns raw flow into an accumulation/distribution read.

6 Fundamental engine. Sector metric registry → Layer 1 regulatory gates → Layer 2 absolute anchors → grade with gate step-down → peer percentiles → DuPont → justified P/B → dividend capacity. Every metric definition carries its own tooltip string in the registry, not in the template.

7 Composite page endpoints. One aggregate per page; add `?fields=` projections for widgets.

8 Frontend. Route shell → dark-first theme bootstrap → page aggregates → per-block empty and stale states → charts last.

9 Auth & billing. Delegate entirely (Clerk/Stripe or equivalent). Keep everything readable signed-out; gate writes and premium depth.

10 Social layer last. It needs a separate store and should degrade to nothing if that store is absent.

Step 8 — Gap analysis vs `nepse_analytics_platform`

Area	Quantitative NEPSE	Your platform
Broker flow	Sankey, top-5 concentration, animated playback, 5D conviction leaders	Broker analytics present; Sankey + concentration + animation are the visible gap
Fundamentals	8 panels, sector-aware, justified P/B, dividend capacity index, two-layer gating	Fundamental analysis exists; the gate/grade model and intrinsic valuation are the gap
Market regime	Scored composite with three tone pillars and named themes	Market insights module — comparable ground, less formalised scoring
Single-stock page	<code>/stock/:ticker</code>	Stock 360 at <code>/stock/<symbol>/</code> — direct analogue
Charting	TradingView Advanced Charts (licensed)	Lighter charting
Social	Full feed, cashtags, sentiment voting, follow graph	None
Auth & paid tiers	Clerk + Stripe pricing table	Django auth, no billing
Portfolio & risk	None at all — deliberately read-only research	Your advantage: holdings import (CSV/Excel/PDF), weekly-return beta, SOP methodology pages, per-user portfolio desk
Data store	DuckDB batch file	MySQL + Django ORM

Strategic read. They win on analytical depth and community; you win on personalisation. The highest-leverage additions for you, in order: (1) broker-flow concentration metrics + Sankey on Stock 360, (2) the two-layer fundamental gate/grade model, (3) justified P/B intrinsic valuation, (4) an explicit data-freshness banner. Numbers 1–3 are analytics you already have most of the raw data for.